

140,000

THE NUMBER OF **PEOPLE EVACUATED**
FROM THE AREA AROUND **THREE MILE**
ISLAND AFTER THE **DISASTER**

An aerial view of the Three Mile Island nuclear power plant, near Harrisburg, Pennsylvania, which is considered the site of the US's worst nuclear accident.

AS THE DECADE DREW TO

A CLOSE, progress in space exploration continued. NASA's **two Voyager space probes** reached their first target, Jupiter, two years after leaving Earth. Voyager 1 made its closest approach in March; Voyager 2 in July. The probes sent back exceptionally clear photographs and data to reveal **new information about Jupiter** and its moons, including the discovery of volcanoes on the innermost moon, Io. In

“FOR EACH OF OUR
ACTIONS THERE ARE ONLY
CONSEQUENCES.”

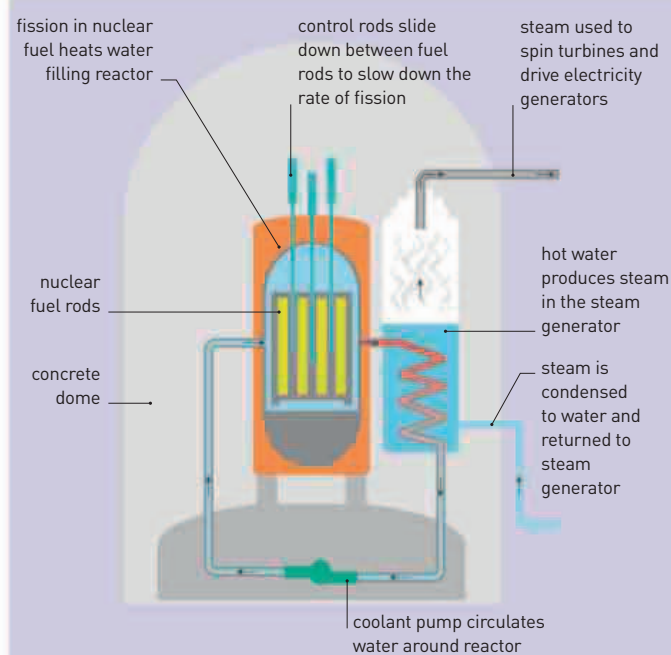
James Lovelock, British biologist, in *Gaia: A New Look at Life on Earth*, 1979

September, **NASA's Pioneer 11** became the first space probe to fly past Saturn, another gas giant. Pioneer 11 came as close as 13,000 miles (21,000 km) to the tops of the planet's deep, dense clouds. At the end of the

year, the European Space Agency launched the first of its highly successful **Ariane rockets**, from French Guyana.

In May, astronomers based at Kitt Peak National Observatory, Arizona, discovered two quasars close to each other in the sky; a quasar (quasi-stellar radio source) is the energetic centre of a distant galaxy. The two quasars appeared so similar that the astronomers concluded they were two views of the same object. This was the **first example of gravitational lensing** (see panel, left): the light from the quasar was bent, as if by a lens, due to the distortion of space-time by a massive galaxy cluster that lies between the quasar and Earth.

On March 28, the US suffered its **worst-ever nuclear power plant accident**. In the early hours of the morning, one of the reactors at the **Three Mile Island nuclear power plant**, Pennsylvania, went into partial meltdown after a valve failed to operate correctly. A large amount of pressurized water circulating

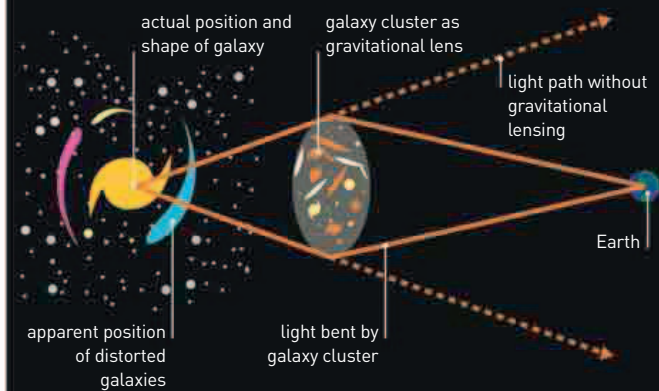


NUCLEAR REACTOR

A nuclear reactor generates electric power by using nuclear fission to heat water. This in turn produces steam, which is used to drive huge turbines. A reactor core contains fuel rods made of radioactive material, and control rods, which limit the rate of fission. There are two separate cooling circuits, in which water circulates to carry heat away from the reactor core.

the reactor core then leaked out into a containment building. Although the reactor was stabilized by the evening, concern over rising radiation levels caused local officials to advise children and pregnant

women within an 5 mile- (8 km-) radius to evacuate; two days after the accident, the radius was extended to 20 miles (32 km). Radiation levels around the power plant did not rise significantly, and nuclear



GRAVITATIONAL LENSING

The gravitational field of matter present in space acts as a lens, causing light to be deflected. As a result, people viewing a celestial object from Earth may see multiple distorted images of it. This phenomenon, known as gravitational lensing, was first proposed in 1936 by German-American physicist Albert Einstein. The first example of gravitational lensing was discovered in 1979, when astronomers observed two views of the same quasar.

January
Bell Laboratories
releases **Version 7**
of the **UNIX** computer
operating system

March 5 Space
probe **Voyager 1**
makes its **closest**
approach to **Jupiter**

March 28
Meltdown at the
Three Mile Island
nuclear power plant

May 31 Astronomers
publish first evidence of
gravitational lensing

June 12 Human-
powered airplane
Gossamer Albatross
crosses the **English**
Channel, winning
the **Kremer Prize**
for the second time

July 1 **Sony**
introduces the
Walkman portable
audio cassette player

July 9 Space probe
Voyager 2 makes its
closest approach
to **Jupiter**

September 1 **NASA's**
Pioneer 11 becomes
the first spacecraft
to fly past **Saturn**